

Lab 14b: Grading the Guess

84-355 Democracy's Data | BISG against the truth, Houston County, Georgia

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Everything here runs as given. Your job is to read what comes out and say what it means.

The question

This morning you learned how BISG works: take a surname, take a place, combine them, and out comes a probability that a voter is Black, or white, or Hispanic. Courts use it. Expert witnesses use it. The Voting Rights Act cases you read for Week 13 rest on it.

And normally you can never check it, because the whole point is that you do not know the answer. That is what makes this an *inference* rather than a measurement.

Today we check it. Georgia is one of a handful of states that records race on the voter registration form — collected for Voting Rights Act compliance, not for anybody's research. So for Houston County we have almost fifty thousand people whose race the method is trying to guess and whose race we already know.

You are about to find out how good the guess is. It is not what the overall number will tell you.

Part 1 — The data

```
v <- read.csv("data/houston_voters.csv", colClasses = c(GEOID20 = "character"))
g <- read.csv("data/houston_blocks.csv", colClasses = c(GEOID20 = "character"))
sn <- read.csv("../surnames/data/census_surnames.csv", stringsAsFactors = FALSE)

str(v)

#> 'data.frame': 49771 obs. of 3 variables:
#> $ surname: chr "AALTONEN" "AARON" "AARON" "AARON" ...
#> $ race : chr "white" "asian" "black" "white" ...
#> $ GEOID20: chr "131530211253020" "131530212041012" "131530203021001" "131530202003009" ...

sort(table(v$race), decreasing = TRUE)

#>
#> white black hispanic asian aian
#> 25333 17701 3201 2300 1236
```

Three columns: a surname, the race the voter reported, and the census block they live in. No names, no addresses — the model only ever gets the surname and the block, so that is all the file carries.

```
nrow(g)
```

```
#> [1] 3269
```

```
colSums(g[, -1])
```

```
#>   white   black hispanic   asian   aian  
#> 86211  56520   11807    5029    339
```

The county is 86,211 white residents and 56,520 Black. That second file is the **geographic prior**: for each of 3,269 blocks, who lives there.

```
R <- c("white", "black", "hispanic", "asian", "aian")  
sc <- c(white = "pctwhite", black = "pctblack", hispanic = "pcthispanic",  
       asian = "pctapi",   aian = "pctaian")  
  
i <- match(v$surname, sn$name)  
cat("surnames found in the Census list:", sum(!is.na(i)),  
    sprintf("(%.1f%%)\n", 100 * mean(!is.na(i))))
```

```
#> surnames found in the Census list: 46031 (92.5%)
```

```
v <- v[!is.na(i), ]  
S <- as.matrix(sn[i[!is.na(i)], sc]); S[is.na(S)] <- 0  
keep <- rowSums(S) > 0  
v <- v[keep, ]; S <- S[keep, , drop = FALSE]  
S <- S / rowSums(S)           # P(race | surname)
```

Part 2 — The surname alone

```
sur <- R[max.col(S, ties.method = "first")]
```

```
cat("SURNAME ONLY\n")
```

```
#> SURNAME ONLY
```

```
cat(sprintf(" overall accuracy: %.1f%%\n\n", 100 * mean(sur == v$race)))
```

```
#> overall accuracy: 61.3%
```

```
for (r in R)  
  cat(sprintf(" %-9s n = %6d correct %.1f%%\n",  
             r, sum(v$race == r), 100 * mean(sur[v$race == r] == r)))
```

```
#>  white      n = 23703  correct 94.7%
#>  black      n = 16852  correct 16.6%
#>  hispanic   n =  2319  correct 71.4%
#>  asian      n =  2038  correct 64.0%
#>  aian       n =  1119  correct  0.2%
```

Stop and look at that before going on. Overall accuracy is 61.3%, which sounds like a method that works. Now read down the column.

It gets white voters right 94.7% of the time and Black voters right **16.6%** of the time. The overall figure is not a summary of how the method performs — it is a summary of how many white people live in Houston County. A method that simply guessed “white” for everybody would score 51.5%.

Part 3 — Adding the place

Surname alone is not BISG. The method combines the surname with where the person lives: if a name is ambiguous but the block is 90% Black, the block should carry the inference.

```
P <- as.matrix(g[, R]); rownames(P) <- g$GEOID20
M <- sweep(P, 2, colSums(P), "/")[v$GEOID20, , drop = FALSE]  # P(block | race)

# Two places the method has no answer, and what we do about it.
empty <- rowSums(M) == 0  # block with no recorded population
M[empty, ] <- matrix(colSums(P) / sum(P), sum(empty), 5, byrow = TRUE)

post <- S * M  # P(race | surname) x P(block | race)
degenerate <- rowSums(post) == 0  # surname and block flatly contradict
post[degenerate, ] <- S[degenerate, ]

post <- post / rowSums(post)
bis <- R[max.col(post, ties.method = "first")]

cat("voters in a block with no census population:", sum(empty), "\n")
```

```
#> voters in a block with no census population: 392
```

```
cat("voters where surname and block cancel out: ", sum(degenerate), "\n")
```

```
#> voters where surname and block cancel out: 8
```

```
cat("BISG: SURNAME + BLOCK\n")
```

```
#> BISG: SURNAME + BLOCK
```

```
cat(sprintf(" overall accuracy: %.1f%%\n\n", 100 * mean(bis == v$race)))
```

```
#> overall accuracy: 67.5%
```

```
for (r in R)
  cat(sprintf(" %-9s n = %6d correct %5.1f%%\n",
              r, sum(v$race == r), 100 * mean(bis[v$race == r] == r)))
```

```
#> white      n = 23703 correct 91.1%
#> black      n = 16852 correct 40.3%
#> hispanic   n = 2319  correct 65.9%
#> asian      n = 2038  correct 55.3%
#> aian       n = 1119  correct  0.4%
```

Overall accuracy rises from 61.3% to 67.5%. That is the number a methods section would report.

Notice the two lines of housekeeping in that chunk, because they are not housekeeping. Some 392 voters live in a block the census records as having nobody in it — new construction, mostly — so the geographic term is empty and we fall back to the county. And for **8 voters the method simply has no answer**: their surname is 100% one race and their block has zero residents of that race, so the multiplication gives zero for every category. BISG is undefined for them.

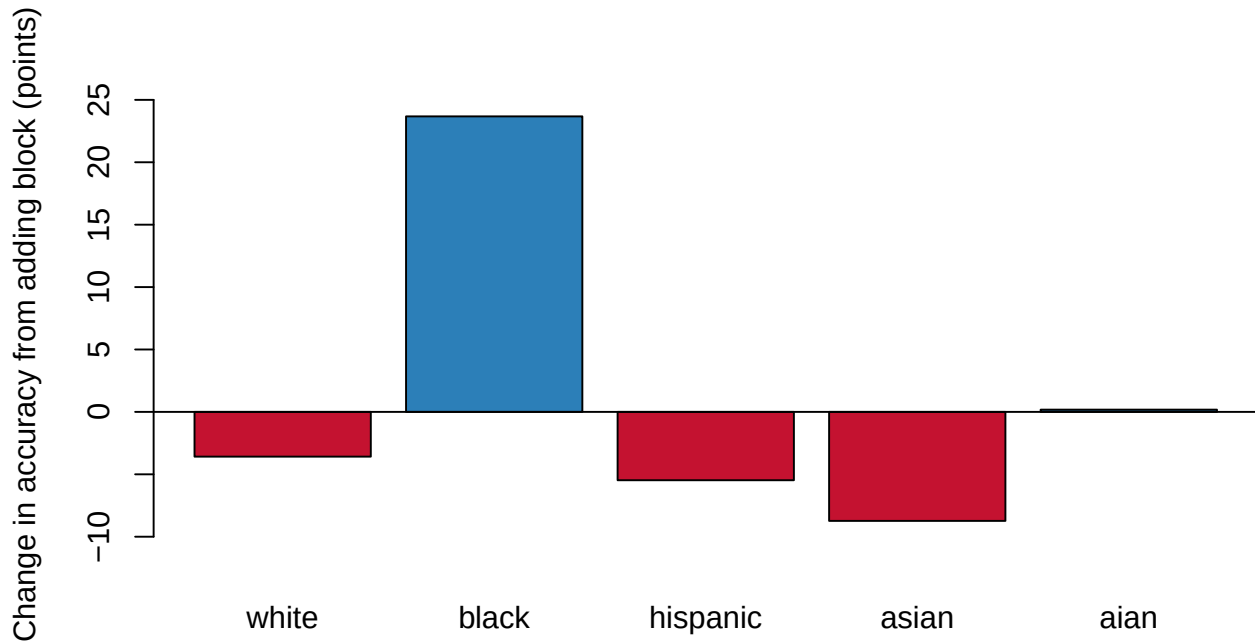
Without those two lines R would not have complained. `max.col` picks the first column when every value is zero, so those 400 people would have been silently predicted **white**, and 158 of them are Black. A quiet default is worse than an error.

Part 4 — Who it helps, and who it costs

```
chg <- sapply(R, function(r)
  100 * (mean(bis[v$race == r] == r) - mean(sur[v$race == r] == r)))
round(chg, 1)
```

```
#> white black hispanic asian aian
#> -3.6 23.7 -5.5 -8.7 0.2
```

```
barplot(chg, col = ifelse(chg > 0, "#2c7fb8", "#C41230"),
        ylab = "Change in accuracy from adding block (points)",
        ylim = c(-12, 26))
abline(h = 0)
```



Adding geography is not a free improvement. It is a trade:

- **Black voters: +23.7 points.** From 16.6% to 40.3%. This is the method working, and it works because Black residents of Houston County are geographically concentrated, so the block genuinely carries information.
- **Asian voters: -8.7 points. Hispanic: -5.5.** The block makes them *worse*. These populations are spread thinly across the county, so no block is distinctively Asian or Hispanic. The geographic term drags the estimate toward the local majority and overrides a surname signal that was already decent.
- **White voters: -3.6 points,** for the same reason in reverse.

The overall +6.2 hides all of this. One group gained twenty-four points, two groups lost.

And the number that should stay with you: even with everything the method has, BISG identifies Black voters correctly **40.3% of the time**. It is wrong about them more often than it is right — in a county that is 35.3% Black, on a question that Voting Rights Act litigation turns on.

Part 5 — Your turn

Change **one** thing, re-run, and write about what changed. Pick one.

Option A — Weight the classes. `max.col` picks the largest posterior, which favours big groups. Instead of the winner, look at the *probability* BISG assigns to the truth: `mean(post[cbind(1:nrow(post), match(v$race, R))]/ rowSums(post))`. Compare that across groups. Is the method badly calibrated, or just badly decided?

Option B — A different geography. Replace the block prior with a county-level one — set every row of `M` to `colSums(P)/sum(P)` — so geography carries no information. What does accuracy fall to, and how close is that to Part 2? What does the answer say about how much of BISG is really the surname?

Option C — Only the ambiguous names. Restrict to voters whose surname gives no group above 60% (`keep <- apply(S, 1, max) < 0.6`) and re-run both methods. These are the people the method is supposed to be *for*. Does it do better or worse on them?

Option D — Your own question. 46,031 voters, 2,123 blocks.

What this lab cannot tell you

- **One county.** Houston County’s population is 53.9% white and 35.3% Black, and moderately segregated. Somewhere more integrated, the block would carry less; somewhere more segregated, more. Do not generalise the numbers, only the shape.
- **The race on the form is also a measurement.** A voter picks one box. People of two races, and Hispanic people of any race, are being flattened by the registration form exactly as Week 12 showed the census flattening them.
- **“Any-part Black.”** We counted Black residents as Black alone *or in combination*, which is Voting Rights Act practice. The stricter definition — Black alone, not Hispanic — gives 51,992 for this county instead of 56,520, a difference of 4,528 people. Every number above would shift.
- **The surname list is from 2010.** The county is not.

On the discussion board

By **Friday, 11:59 p.m.** No length requirement; a sentence or two is enough, so long as it shows you engaged with what we did. Any *one* of the usual four.

Three that would make good posts:

- A method that is right 67.5% of the time overall and 40.3% of the time for Black voters is used in federal voting-rights cases. Is that acceptable? Say what would change your mind.
- Adding geography helped one group and hurt two. Would you use it?
- American Indian voters — 1,119 of them — are identified correctly 0.2% of the time by the surname alone and 0.4% of the time by BISG. What happens to a group the method cannot see?

Where the data came from

Houston County, Georgia voter registration with self-reported race, and the geocoded block assignment, from working data for the 2026 Houston County redistricting matter. Block population by race is 2020 Census PL 94-171 / DHC tables. Surname probabilities are the 2010 Census surname list you used this morning.

Rows where race had been filled in by BISG rather than reported by the voter were dropped before anything else. Scoring BISG against BISG would have produced an excellent and completely meaningless result. The check is in `data/build-data.R`.